

Streamlining IT Procurement

Automating Standard Laptop Orders with Flow Designer — Comprehensive Documentation

1. Brainstorming & Ideation Phase

1.1 Industry Context & Executive Summary: In modern enterprise environments, IT Asset Management (ITAM) and Hardware Procurement are crucial functions that directly impact employee productivity and onboarding timeline. Historically, employees and departmental managers request hardware through manual channels such as email chains, paper requisition forms, or unstructured ticketing channels. This unautomated approach creates operational bottlenecks, extends hardware fulfillment cycles, and increases operational costs. By leveraging **ServiceNow Flow Designer**, this project transforms the standard laptop procurement model into a fully automated, transparent, and low-touch digital workflow.

1.2 Core Problem Statements:

- **Prolonged SLA & Onboarding Delays:** Manual routing of procurement approvals frequently stalls in manager inboxes.
- **Lack of Process Standardization:** Inconsistent collection of user specifications and shipping addresses causes administrative re-work.
- **Inventory Mismanagement:** Absence of real-time integration with Asset Management databases (`alm_hardware`) leads to duplicate purchases.
- **Audit & Compliance Deficits:** Unstructured approvals via email fail to meet enterprise compliance and audit trail requirements.

1.3 Proposed Solution & System Capability: Implement an end-to-end automated Service Catalog Item—**Standard Laptop Order**—backed by a robust Flow Designer workflow engine in ServiceNow covering Self-Service Requisition, Dynamic Multi-Level Approvals, Automated Fulfillment Operations, and Lifecycle Tracking.

2. Requirement Analysis Phase

2.1 Functional Requirements:

Req ID	Module / Area	Feature Description	Acceptance Criteria
FR-01	Service Catalog	Catalog Item Creation	Interface allows selection of laptop tier, justification, and delivery address.
FR-02	Workflow Engine	Approval Routing	Request automatically queries user profile to identify line manager and route approval.
FR-03	Exception Handling	Rejection Logic	If rejected, update RITM state to 'Closed Incomplete' and log rejection reason.
FR-04	Fulfillment	Task Generation	Auto-create sequential <code>sc_task</code> records for Procurement, Imaging, and Shipping teams.

Req ID	Module / Area	Feature Description	Acceptance Criteria
FR-05	Notifications	Status Communication	Send triggered email notifications upon Request Creation, Approval, and Dispatch.

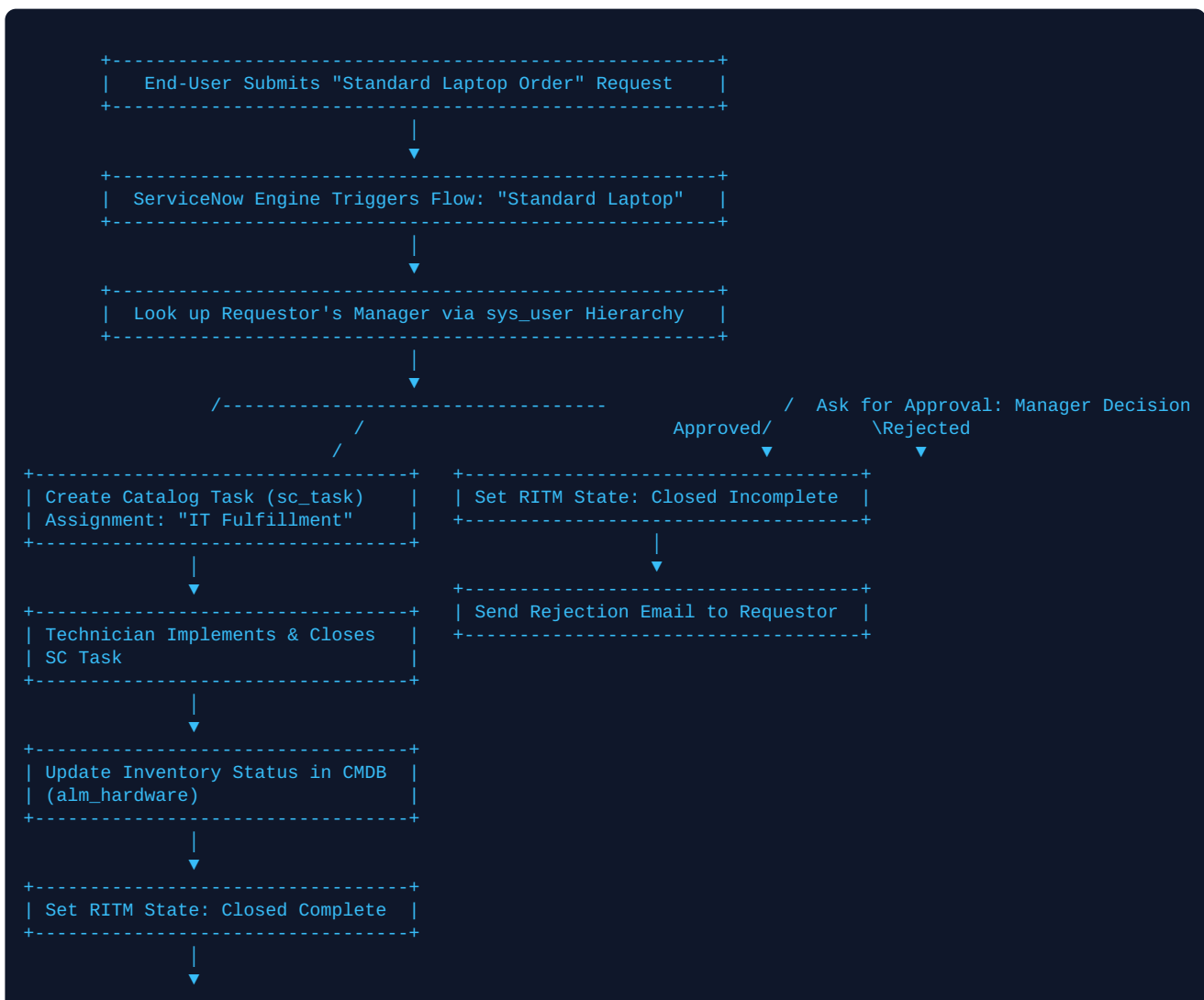
2.2 Non-Functional Requirements:

- **System Scalability:** Workflow architecture designed to process >1,000 concurrent catalog requests without degradation.
- **Execution Efficiency:** Workflow trigger to approval generation execution latency < 1.5 seconds.
- **Role-Based Access Control (RBAC):** Restrict catalog item visibility based on user roles (`itil` , `employee`).
- **Audit Trail:** Maintain an immutable record of all approval timestamps and decisions in `sys_audit` .

2.3 Technology Stack & Target Tables: ServiceNow Developer Instance (San Diego/Utah/Washington), Service Catalog, Flow Designer, ITSM. Target Tables: `sc_request` , `sc_req_item` , `sc_task` , `sys_user` , `alm_hardware` .

3. Project Design Phase

3.1 Workflow Architecture Diagram:



```
+-----+
| Send Delivery Confirmation Email |
+-----+
```

3.2 Catalog Variable Schema Design:

- `requested_for` (Reference -> `sys_user`, Mandatory)
- `laptop_model` (Select Box: Enterprise Tier 1 [16GB/512GB], Developer Tier 2 [32GB/1TB])
- `business_justification` (Multi Line Text, Mandatory)
- `delivery_location` (Reference -> `cmn_location`, Mandatory)
- `contact_number` (String, Optional)

4. Project Planning Phase

Milestone	Sub-Tasks	Estimated Effort	Deliverables
Phase 1: Analysis	Catalog variables definition, process mapping	Day 1 - 2	Requirement Sign-off
Phase 2: Catalog Build	Portal form layout design, dictionary overrides	Day 3	Active Catalog Item
Phase 3: Flow Design	Triggers, Manager approvals, Task creation logic	Day 4 - 5	Flow Designer Artifacts
Phase 4: Notifications	HTML Email templates, Notification triggers	Day 6	Configured System Mailers
Phase 5: Quality Testing	UAT scenarios, Regression testing, Bug fixing	Day 7	Test Results Matrix
Phase 6: Deployment	Update Set commit, Demo video, Handover	Day 8	Final Submission

5. Project Development Phase

Step 1: Catalog Item Configuration

- Created Catalog Item: `Standard Laptop Order` under `Service Catalog > Hardware`.
- Added Catalog Variables: `requested_for`, `laptop_model`, `business_justification`, and `delivery_location`.

Step 2: Flow Designer Logic Development

- **Trigger:** Service Catalog Record Created (`sc_req_item`).
- **Action 1:** Look up `sys_user` record of the `Requested For` user to extract the Manager field.
- **Action 2 (Ask For Approval):** Route approval to Manager.
- **If Approved:** Create Catalog Task (`sc_task`) assigned to `IT Asset Management`, send approval email, update state to `Closed Complete` upon task completion.
- **If Rejected:** Update RITM State to `Closed Incomplete` and dispatch email with rejection comments.

6. Project Testing Phase

Test Case ID	Test Objective	Execution Steps	Expected Outcome	Status
TC_PROC_01	Standard Order Submission	Fill catalog item form and click submit.	RITM generated with state 'Awaiting Approval'.	PASS
TC_PROC_02	Approval Request Trigger	Submit order with assigned manager profile.	Approval record created under Manager portal.	PASS
TC_PROC_03	Approval Rejection Route	Manager rejects request with reason text.	Workflow skips task creation; RITM closed as incomplete.	PASS
TC_PROC_04	Task Auto-Assignment	Manager approves order.	Catalog Task generated & assigned to IT Asset group.	PASS
TC_PROC_05	Notification Validation	Trigger approval and fulfillment events.	HTML notifications dispatched to user email address.	PASS

7. Project Documentation Phase

System README Setup Summary:

This repository contains ServiceNow configuration files, update sets, and architectural designs for automating hardware procurement workflows. Deploy by logging into ServiceNow as Administrator, importing the XML Update Set via `System Update Sets > Retrieved Update Sets`, committing the set, and opening `Service Catalog > Standard Laptop Order` for testing.

8. Project Demonstration Phase

Demo Video Presentation Script:

0:00 - 0:30 (Title & Intro): "Hello everyone. My name is [Your Name], and today I am presenting my project: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer."

0:30 - 1:15 (Purpose & Problem): "In traditional corporate environments, ordering a laptop involves scattered emails and manual task creation. The goal of this project is to eliminate manual intervention by building an end-to-end automated workflow using ServiceNow Flow Designer."

1:15 - 2:45 (Live Demo): Show Catalog Form submission on portal, Flow Designer execution, Manager approval portal, automated SC Task generation for IT Asset Management, and auto-notification emails.

2:45 - 3:30 (Conclusion): "Upon completing the IT fulfillment task, the workflow automatically updates the RITM state to 'Closed Complete' and logs asset details, reducing procurement SLA from days to hours. Thank you."